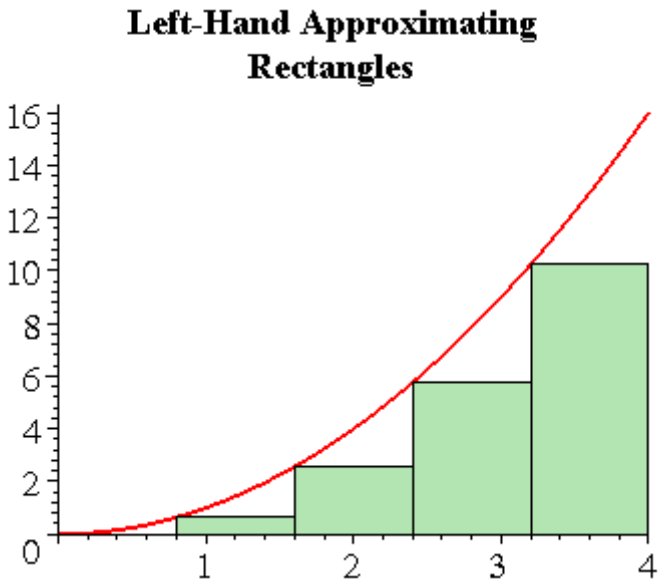
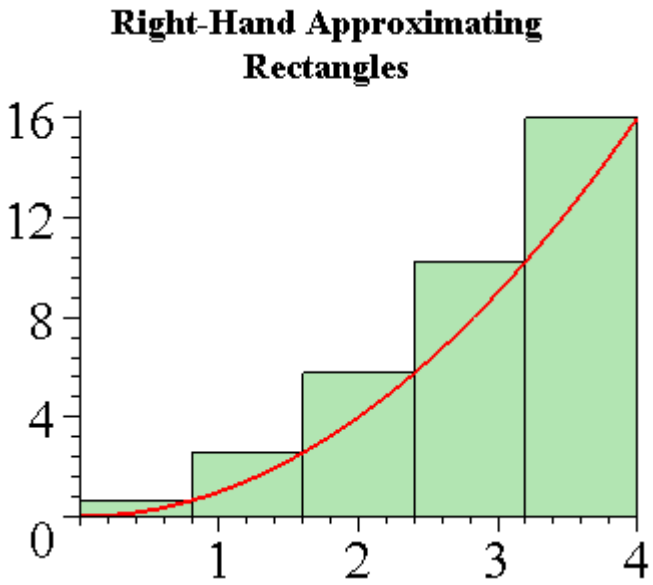
Sec 8.3 Integral and Comparison tests

Series are very important in engineering as they are often used to model real world problems. It is useful to have methods for determining weather or not a series converges.

One such test is the integral test. You might recall from your first semester of calculus the left and right Riemann sums.



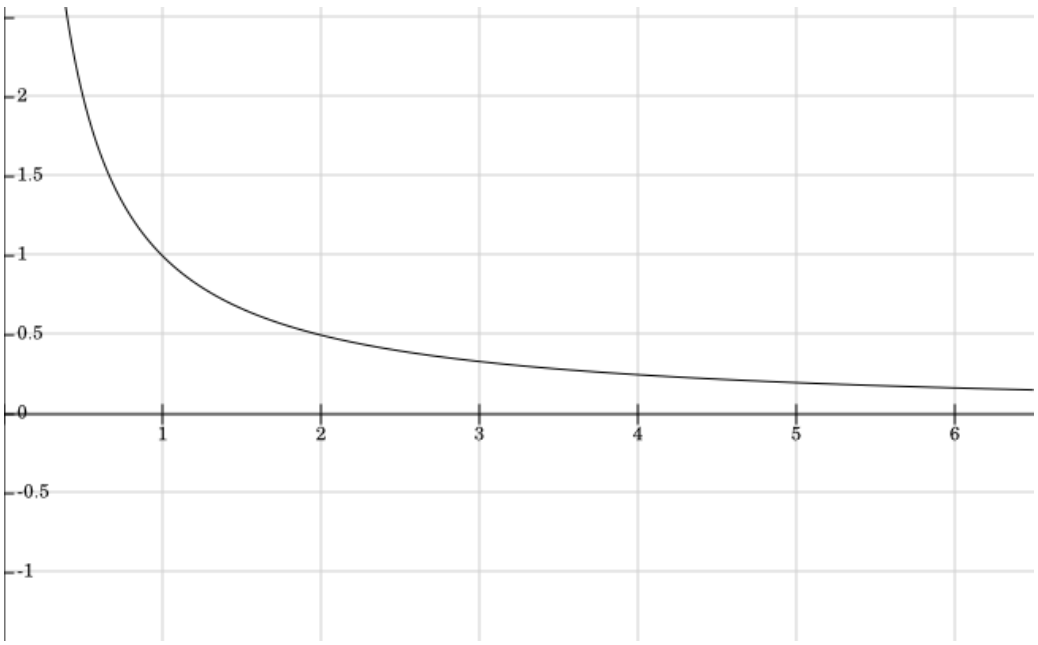


The left hand sum underestimates the area

The right hand sum over estimates the area

Consider

$$\sum_{n=1}^{\infty} \frac{1}{n}$$



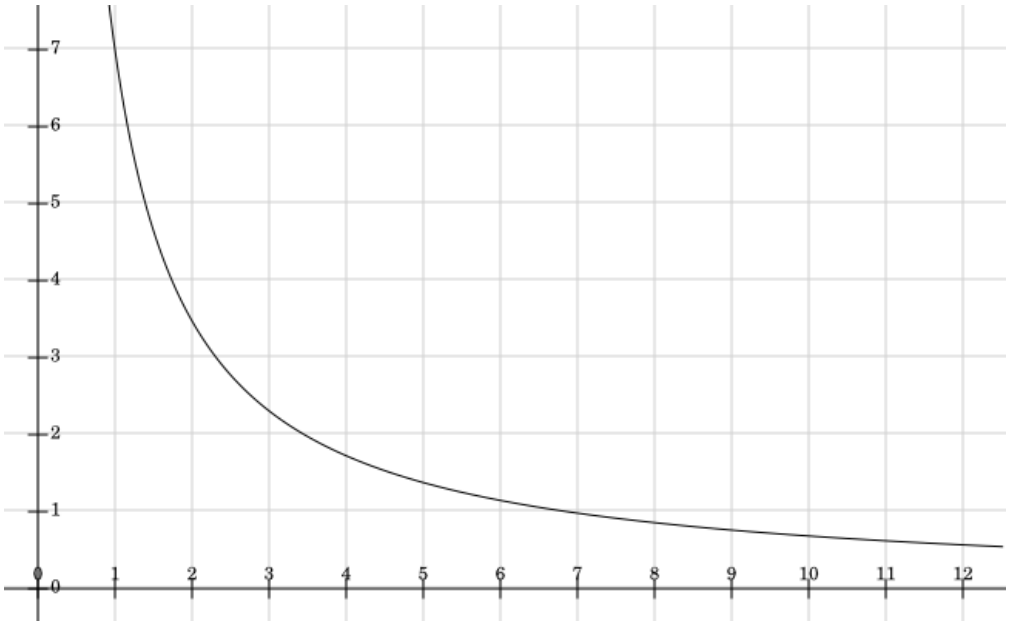
So the with $\Delta x = 1$ the Riemann sums are the same as the series!

Sec 8.3 Integral and Comparison tests

Lets Suppose we wonder whether or not the series

$$\sum_{i=1}^{\infty} \frac{7}{i^{1.01}}$$

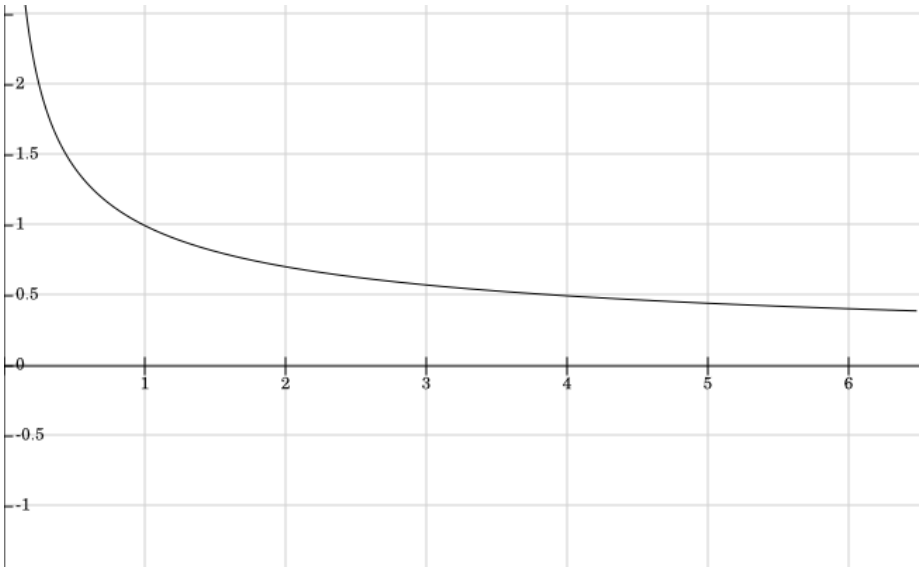
Converges



Sec 8.3 Integral and Comparison tests

Does the series converge?

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$



Sec 8.3 Integral and Comparison tests

The Integral Test Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x)\,dx$ is convergent. In other words:

(a) If $\int_1^{\infty} f(x)\,dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.

(b) If $\int_1^{\infty} f(x)\,dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

Ex. Determine whether the series converges

$$\sum_{i=1}^{\infty} \frac{1}{nln(n)}$$

Sec 8.3 Integral and Comparison tests

Consider the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ For what values of p does this series converge?

1 The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leq 1$.

Sec 8.3 Integral and Comparison tests

The Comparison Test Suppose that $\Sigma\, a_n$ and $\Sigma\, b_n$ are series with positive terms.

(a) If $\Sigma\, b_n$ is convergent and $a_n \leq b_n$ for all n, then $\Sigma\, a_n$ is also convergent.

(b) If $\Sigma\, b_n$ is divergent and $a_n \geq b_n$ for all n, then $\Sigma\, a_n$ is also divergent.

Ex. Show that the series $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n^5 + 3n^2 + 1}$ converges

Ex. Show that the series $\displaystyle\sum_{n=0}^{\infty} \frac{\ln(n)}{n}$ diverges

Sec 8.3 Integral and Comparison tests

The Limit Comparison Test Suppose that $\Sigma\, a_n$ and $\Sigma\, b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

Example Does the series $\sum_{n=1}^{\infty} \frac{1}{n^2 - n}$ Converge?

What about the series $\sum_{n=1}^{\infty} \frac{1}{n + \sqrt{n}}$

Sec 8.3 Integral and Comparison tests

What if we want to know how far off a partial sum is from the full infinite sum? We can define the remainder as:

$$R_N = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{N} a_n = \sum_{n=N+1}^{\infty} a_n = s - s_n$$

Then the question is , can we estimate this error?

Let's Think about the integrals…..

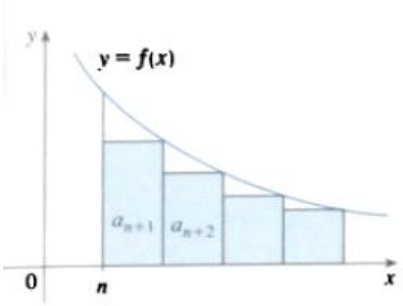


FIGURE 3

$$R_N = \sum_{n=N+1}^{\infty} a_n \leq \int_N^{\infty} a(x)\,dx$$

Note $\Delta x = 1$

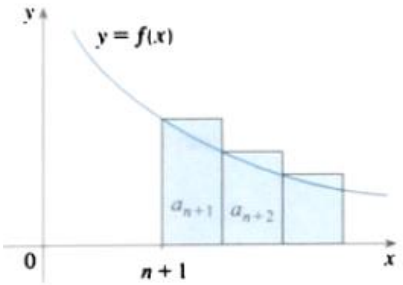


FIGURE 4

$$R_N = \sum_{n=N+1}^{\infty} a_n \geq \int_{N+1}^{\infty} a(x)\,dx$$

3 Remainder Estimate for the Integral Test Suppose $f(k) = a_k$, where f is a continuous, positive, decreasing function for $x \geq n$ and $\Sigma\, a_n$ is convergent. If $R_n = s - s_n$, then

$$\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_n^{\infty} f(x)\,dx$$

Sec 8.3 Integral and Comparison tests

Using this we can get bounds on the actual sum

3 Remainder Estimate for the Integral Test Suppose $f(k) = a_k$, where f is a continuous, positive, decreasing function for $x \geq n$ and $\Sigma\, a_n$ is convergent. If $R_n = s - s_n$, then

$$\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_{n}^{\infty} f(x)\,dx$$

Replacing R_n *with* $s - s_n$ *and adding* s_n*to both sides gives*:

Bounds on an infinite sum!

$$s_n + \int_{n+1}^{\infty} f(x)\,dx \leq s \leq s_n + \int_{n}^{\infty} f(x)\,dx$$

Ex. Estimate the error in approximating the series below only using 10 terms.

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Sec 8.3 Integral and Comparison tests

How many terms do we need to make sure the error in an estimate of the sum below is with in 0.0005?

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Sec 8.3 Integral and Comparison tests

Find bounds for the sum

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Using the 10th partial sum